



AI for Hematological Image Analysis

Building, training, and augmenting
plasma cell detectors with generative
AI

Instructors



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What you'll see in this tutorial

- 01 **Two expert-curated datasets** — real bone-marrow exam images from patients with and without Multiple Myeloma
 - 02 **Cell detection & classification** models
 - 03 **Diagnostic models** — from cell counts to a patient-level decision
 - 04 **Image-processing techniques** for microscopy
 - 05 **Generative AI** for medical image processing
-

What we'll cover today

14:00–14:20	Multiple Myeloma & the diagnostic bottleneck	TALK
14:20–14:45	Hands-on 1 — loading & visualizing the dataset	NOTEBOOK
14:45–15:10	Detecting plasma cells with YOLO + Hands-on 2	TALK • NOTEBOOK
15:10–15:30	The PCMMMD dataset & per-patient diagnosis + Hands-on 3	TALK • NOTEBOOK
15:30–16:00	Break	—
16:00–16:40	Synthetic data augmentation for microscopy	TALK
16:40–17:15	Hands-on 4 — building & testing synthetic data	NOTEBOOK
17:15–17:30	Results, takeaways & wrap-up	DISCUSSION

BEFORE WE BEGIN

Code along with us.

Everything in this tutorial — the data loaders
and the model code — lives in one place.
Open it now and follow along hands-on.

➤ labia.ufba.br/mm-tutorial



SCAN TO OPEN

THE SOURCE

The papers behind the datasets

Both peer-reviewed in the Nature portfolio — scan to read the source.



DATASET · DIAGNOSIS

Scientific Reports, 2024

[nature.com/articles/
s41598-024-61420-9](https://nature.com/articles/s41598-024-61420-9)



DATASET · PCMMD

Scientific Data, 2025

[nature.com/articles/
s41597-025-04459-1](https://nature.com/articles/s41597-025-04459-1)

01

Why this matters

The clinical bottleneck behind Multiple Myeloma diagnosis — and why it's a hard vision problem.

The most incident plasma-cell cancer

More frequent in people over 50 — and diagnosis still leans on the manual myelogram.

176k

new cases worldwide, 2020 (WHO)

117k

deaths worldwide in the same year

10–15%

of all hematological neoplasms
(~1% of all tumors)

~2.6k

cases/year in Brazil, 2013–2019

01 • MOTIVATION – THE CELL

What is a plasma cell?

Effector cells of the B-lymphocyte lineage — they produce and secrete the antibodies behind the humoral immune response.

B lymphocyte → plasmablast → proplasmocyte → **plasma cell**

Diameter **12–18 μm**

Nucleus **dense, eccentric**

Cytoplasm **deeply basophilic**

Tell-tale sign **perinuclear halo**

Markers **CD27 • CD38 • CD138**

What goes wrong in Myeloma

A malignant clone of plasma cells **proliferates out of control and floods the bone marrow** — driving monoclonal-protein secretion (antibody chains in blood or urine) and organ dysfunction.

COMMON SYMPTOMS

Fatigue & weakness

Anemia & weight loss

Kidney dysfunction

Bone pain — spine

Spontaneous fractures

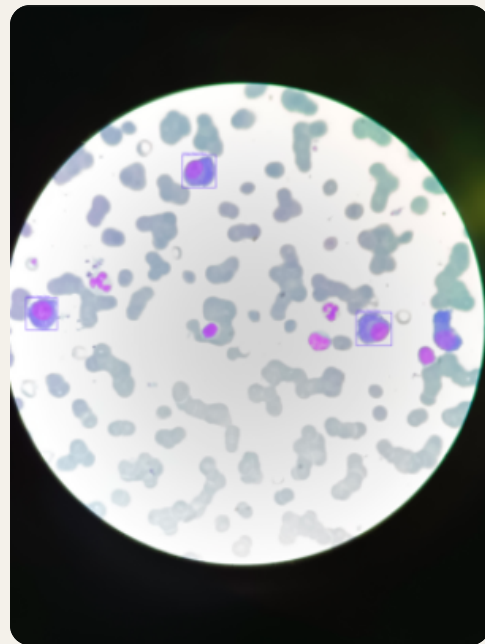
Recurrent infections

01 • MOTIVATION

A cancer of plasma cells in the bone marrow

Multiple Myeloma is a hematological malignancy that originates from a clone of malignant plasma cells infiltrating the bone marrow.

Diagnosis is confirmed on **bone marrow aspirate smears** — stained, then read cell by cell by a specialist.



01 · MOTIVATION — THE DIAGNOSTIC CRITERION

≥10%

clonal plasma cells in the bone marrow signals a positive diagnosis (IMWG criteria).

Assessing that ratio means manually counting **200–500 cells** per slide.

A slow, observer-dependent count

Labor-intensive

Hundreds of cells per slide, by eye. Time and resources that can delay diagnosis and treatment.

Observer-dependent

Accuracy hinges on the expert's attention and experience, introducing inconsistency between readers.

Hard to scale

A shortage of trained specialists limits screening in resource-constrained communities.

A microscope and a smartphone

A low-cost rig can accelerate reads, improve consistency, and enable preliminary screening where specialists are scarce — from sample to labeled data.

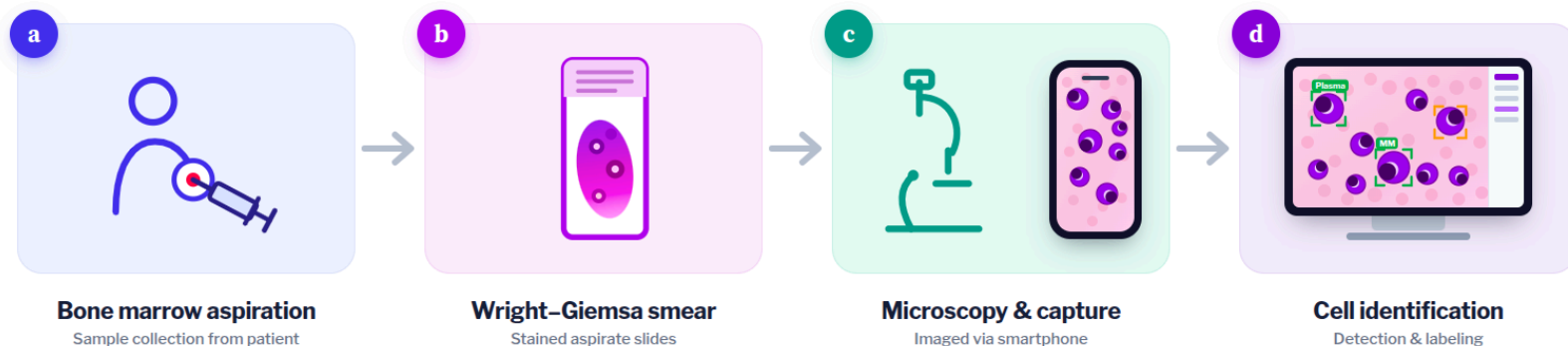


Figure 1. (a) Bone marrow aspirate procedure; (b) Wright–Giemsa stained bone marrow aspirate smear slides from MM patients, analyzed by the oncohematology and immunophenotyping service of the Laboratory of Immunology and Molecular Biology; (c) Observation of stained slides in visible light optical microscopes and image capture by smartphone device; (d) Identification and labeling of detected cells.

HANDS - ON

Let's code!

Loading and visualizing the first dataset.

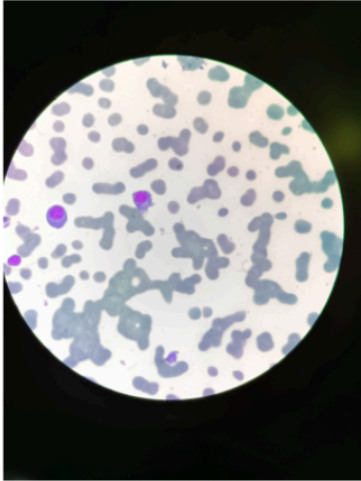
What the notebook does

-
- 01 Install `ultralytics` and download the PCMMMD detection folder
 - 02 Parse YOLO-format labels — one `.txt` per image, normalized boxes
 - 03 Draw boxes over sample slides to sanity-check annotations
 - 04 Plot class balance and bounding-box size distribution
-

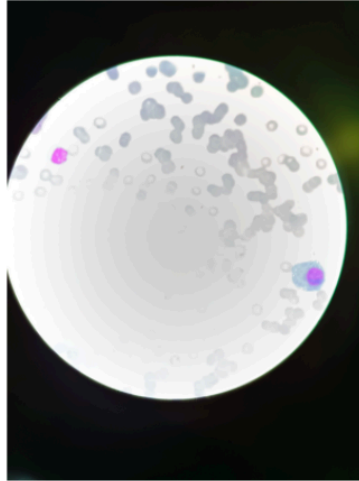
Reading the raw fields

Straight from the loader — circular field of view, pale red-cell background, and bright nucleated cells we'll learn to detect.

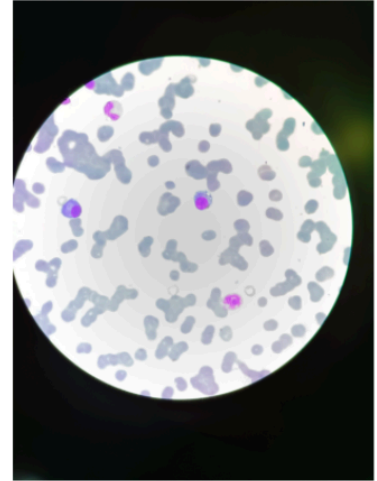
1692243677913.jpg

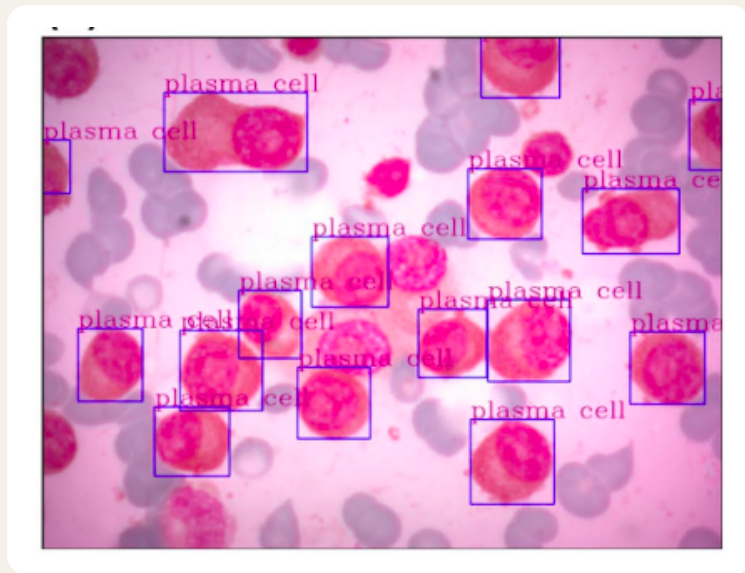


1692243677966.jpg



1692243677995.jpg





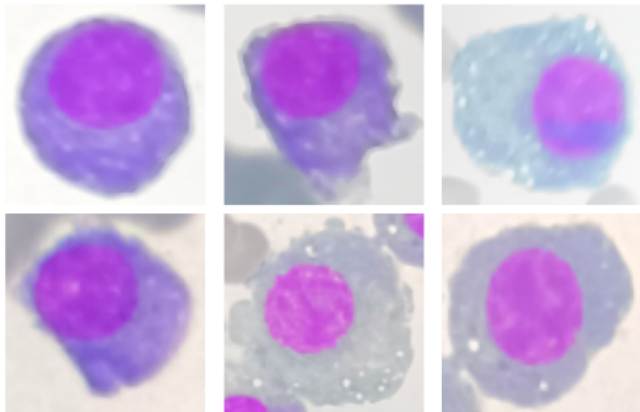
01 • THE LABELS

The same field, labeled

Experts box and classify every nucleated cell — this overlay is the ground truth the detector learns to reproduce.

☐ **Plasma Cell**

Partially-cut cells and stain artifacts are left unlabeled.



One labeled cell at a time

Each cell an expert labels is cropped into its own image — sharing the same morphology: dense eccentric nucleus, basophilic cytoplasm, perinuclear halo.

These crops are the **raw material for training** — the examples our generative models learn to synthesize.

02

How to detect plasma cells

Framing detection as a computer-vision task —
and training a YOLO model to find cells
automatically.

Detection is the core task

Small objects

Cells occupy a small fraction of each patch, amid clutter and morphological variability.

One class

The tutorial focuses on the plasma-cell class for detection and synthesis.

Patient-wise split

Test patients are held out entirely — no leakage, honest generalization.

409 train imgs 103 val 344 test · 10 patients

Training recipe: YOLO26n

2.4M parameters · 5.4 GFLOPs — edge-ready

640 px input · 100 fine-tuning epochs

COCO pretrained → fine-tuned on PCMMD

ProgLoss & STAL — progressive loss balancing and small-target-aware label assignment, tuned for small objects at the scale of plasma cells.

Full rotation + flips (cells have no canonical orientation); photometric jitter reduced, since deconvolution already handles color.

02 · DETECTION — ARCHITECTURE

YOLO26n / Plasma Cell Detector

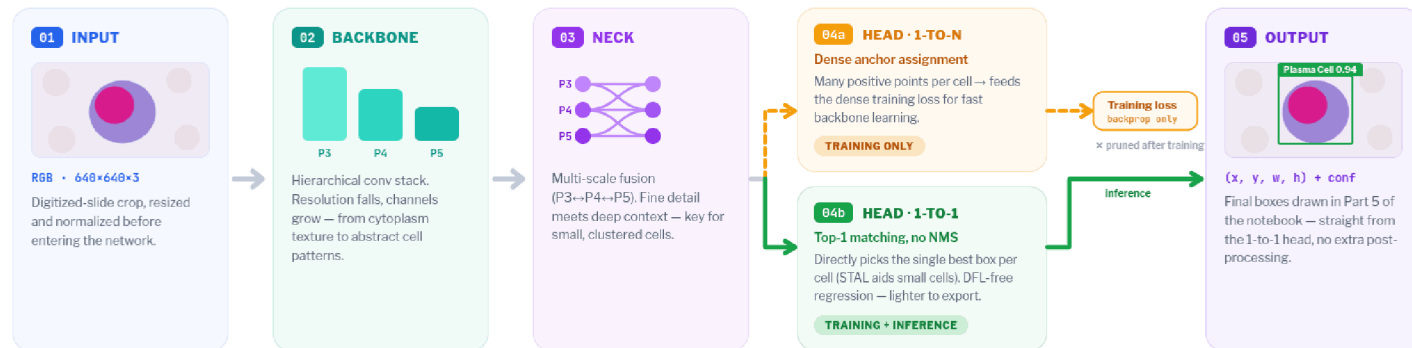
Data flow of the model used in `02_training_yolo.ipynb` to localize plasma cells on slides from Dataset 1 (Andrade et al., 2024).

INITIAL WEIGHT
yolo26n.pt (COCO)

CLASSES
1 — Plasma Cell

INPUT
640×640×3

POST-PROC.
NMS-free

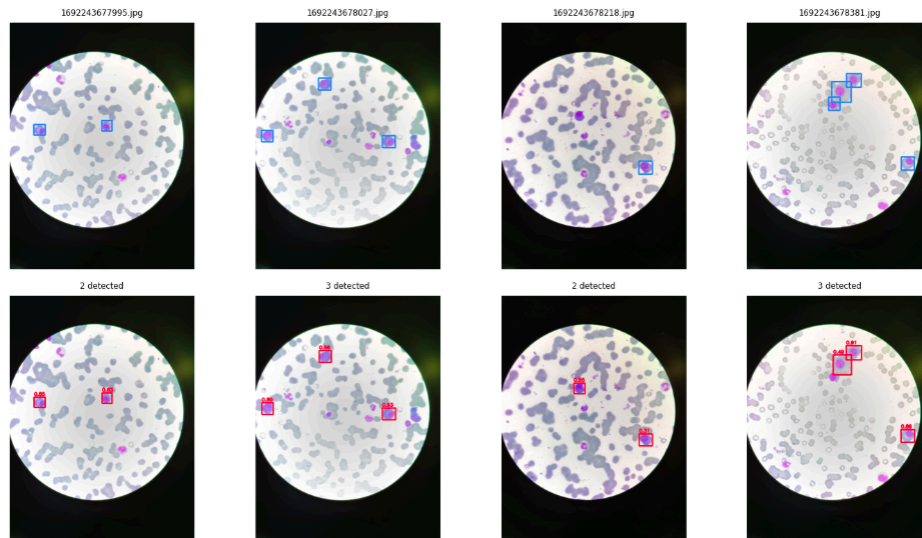


Conceptual sketch, not an official layer diagram. Ultralytics does not publish the exact block/kernel breakdown of YOLO26; the granularity above is simplified for didactic purposes, based on the model's public documentation.

SOURCES

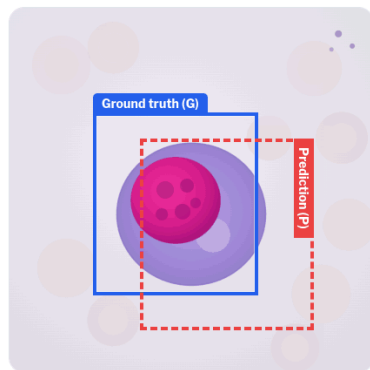
Ultralytics · YOLO26 End-to-End NMS-Free Detection
Ultralytics · YOLO26 Models for Real-Time Edge AI
arXiv 2606.03748 · YOLO26: Unified Real-Time End-to-End Vision Models

What you should see

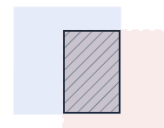


Intersection over Union

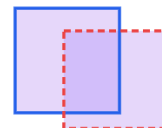
Ground truth (G) vs. prediction (P) — the overlap ratio behind every detection metric.



(a) Bounding boxes on ROI
Specialist annotation vs. model output



(b) Intersection $G \cap P$
Overlap shared by both boxes



(c) Union $G \cup P$
All area covered by either box

$$IOU = \frac{\text{Intersection } G \cap P}{\text{Union } G \cup P} \quad (1)$$

Figure 4. Bounding boxes illustrating a Region of Interest (ROI) defined by specialists (**ground truth**) and a classification output (**prediction**). **(b)** The intersection is the area shared by both boxes; **(c)** the union is the total area covered by either box. The Intersection over Union (IOU) is their ratio, as defined in Eq. (1).

The metrics you'll read

$$\text{IoU} = \frac{\text{area}(G \cap P)}{\text{area}(G \cup P)}$$

Overlap between ground-truth box G and prediction P — the basis of every score below.

Precision

Of the boxes predicted, how many are correct — $TP / (TP + FP)$.

Recall

Of the real cells, how many were found — $TP / (TP + FN)$.

mAP50

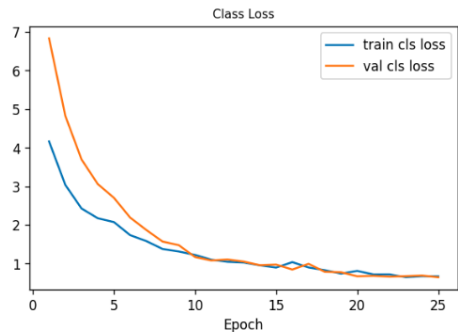
Average precision at a lenient IoU threshold of 0.50.

mAP50-95

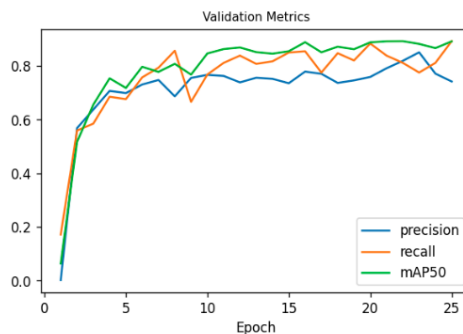
Averaged over IoU 0.50→0.95 — the headline metric, rewarding tight boxes.

What we track during training

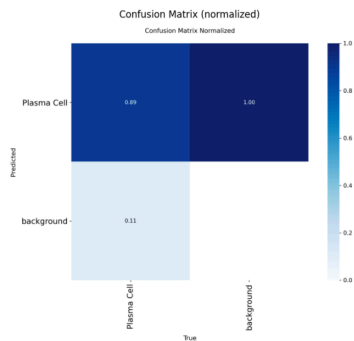
Class loss train vs. val, per epoch



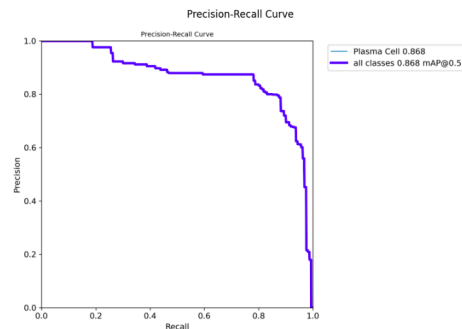
Validation metrics precision · recall · mAP50



Confusion matrix normalized, per class



Precision-Recall mAP@50 = 0.868



Let's code!

Training a model for cell detection.

What the notebook does

-
- 01 Build a **patient-wise split** — test patients never seen in training — and write `data.yaml`
 - 02 Load `YOLO26n` pretrained on COCO — single *Plasma Cell* class
 - 03 Fine-tune — `model.train(imgsz=640, epochs=100)`, rotations + flips
 - 04 Evaluate on held-out patients (mAP50 / mAP50-95) and visualize predicted boxes
-

03

The PCMMD dataset

Plasma Cells for Multiple Myeloma Diagnosis —
a public, expert-annotated benchmark.

How the dataset was built

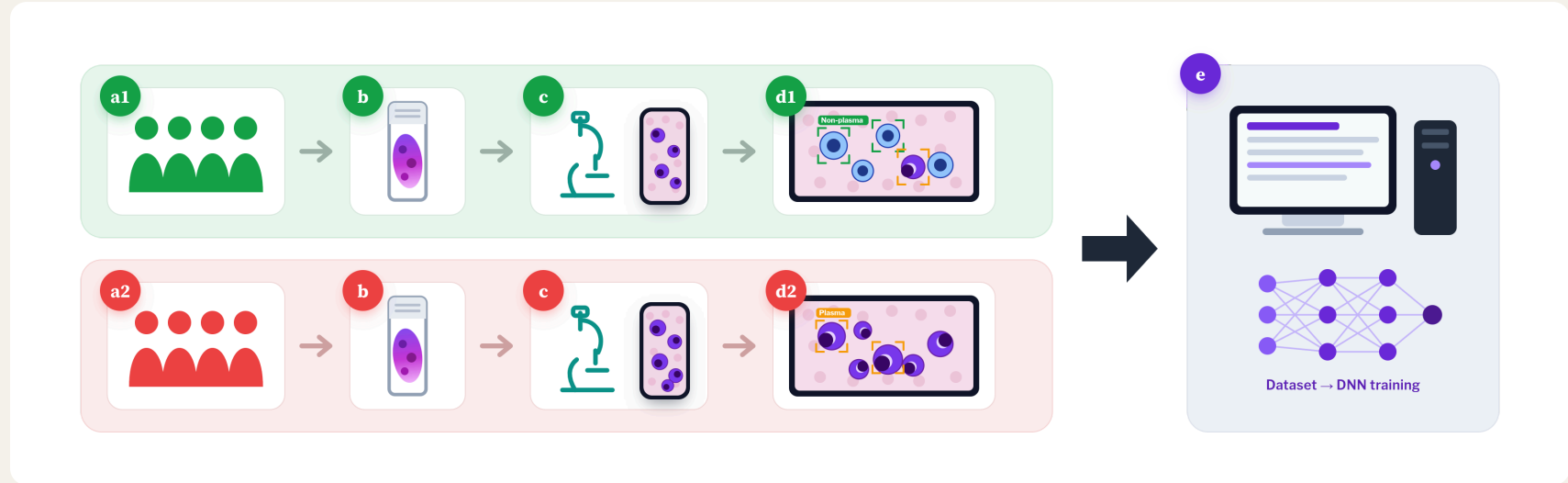


Fig. 1 (a1) Bone marrow aspirate procedure of patients without MM; (a2) Bone marrow aspirate procedure of patients with MM; (b) Wright–Giemsa (SIGMA-ALDRICH, MERCK) stained bone marrow aspirate smear slides, analyzed by the oncohematology and immunophenotyping service of the Labimuno; (c) Observation of stained slides in visible light optical microscopes and image capture by smartphone device; (d1 and d2) Identification, labeling and validation of detected non-plasma and plasma cells from patients with and without MM; and (e) Dataset organization to be later trained by DNN models.

What's inside PCMMD

5,000+

expert-labeled plasma & non-plasma cells

2

classes — plasma vs non-plasma

10

patients with diagnostic ground truth

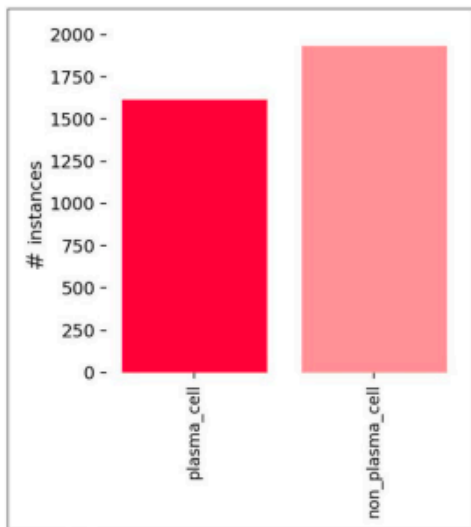
CC BY

open license · Mendeley Data + GitHub

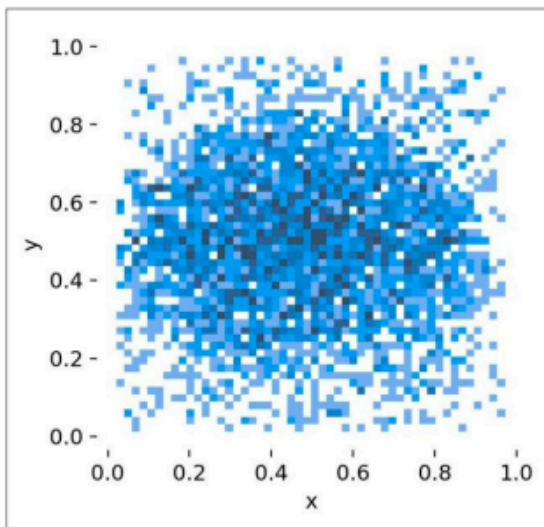
Two folders — **detection** (slides + YOLO labels) and **segmentation** (single-cell crops + masks) — support both detectors and generative models.

Label & bounding-box distributions

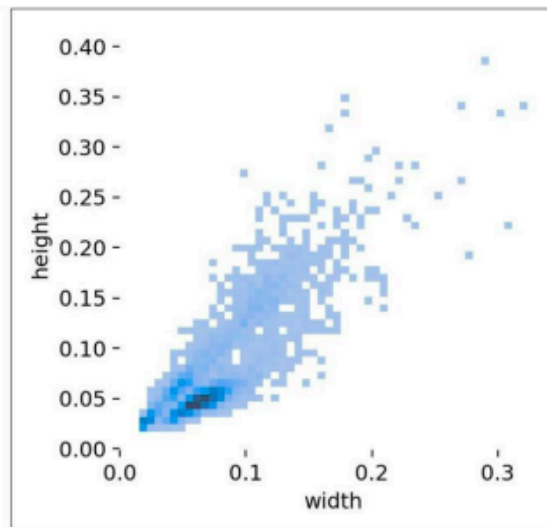
(a) counts per class — mild imbalance · (b) cell positions across the field · (c) box shape: small and near-square.



(a)



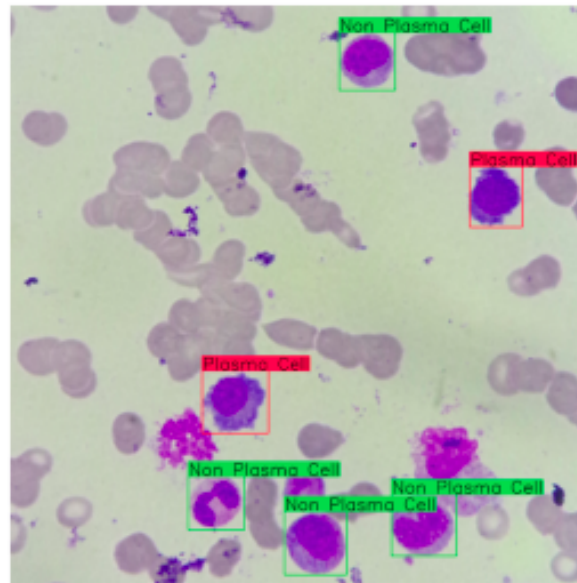
(b)



(c)

Annotated by experts, cross-validated

- Two expert groups label separately and validate each other's boxes
- Boxes follow cytomorphology under Wright–Giemsa staining
- **Plasma** cells in red, **non-plasma** in green
- Partially-cut cells and stain artifacts are left unlabeled



A YOLOv8 detector, already validated

84%

F1-score identifying plasma vs non-plasma

F1 = harmonic mean of precision & recall

>90%

mAP50 on the detection task

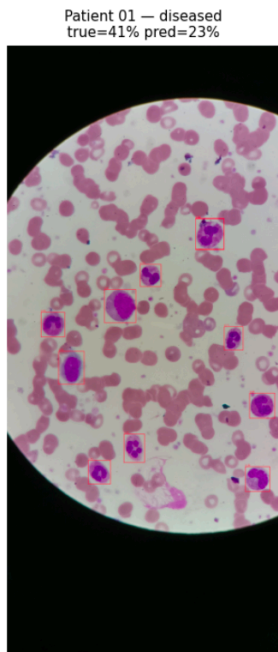
9/10

patient diagnoses agree with experts

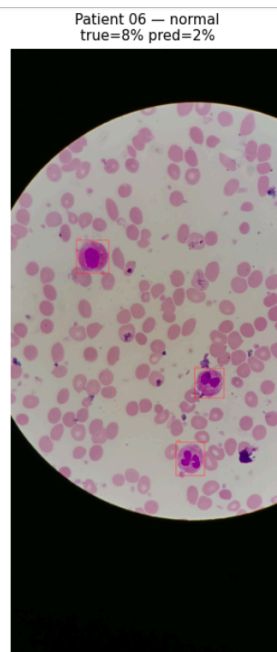
Transfer-learned from COCO, then fine-tuned on PCMMD — the foundation the rest of this tutorial extends.

Diseased vs. normal, side by side

Patient 01 — **diseased** true=41% · pred=23%



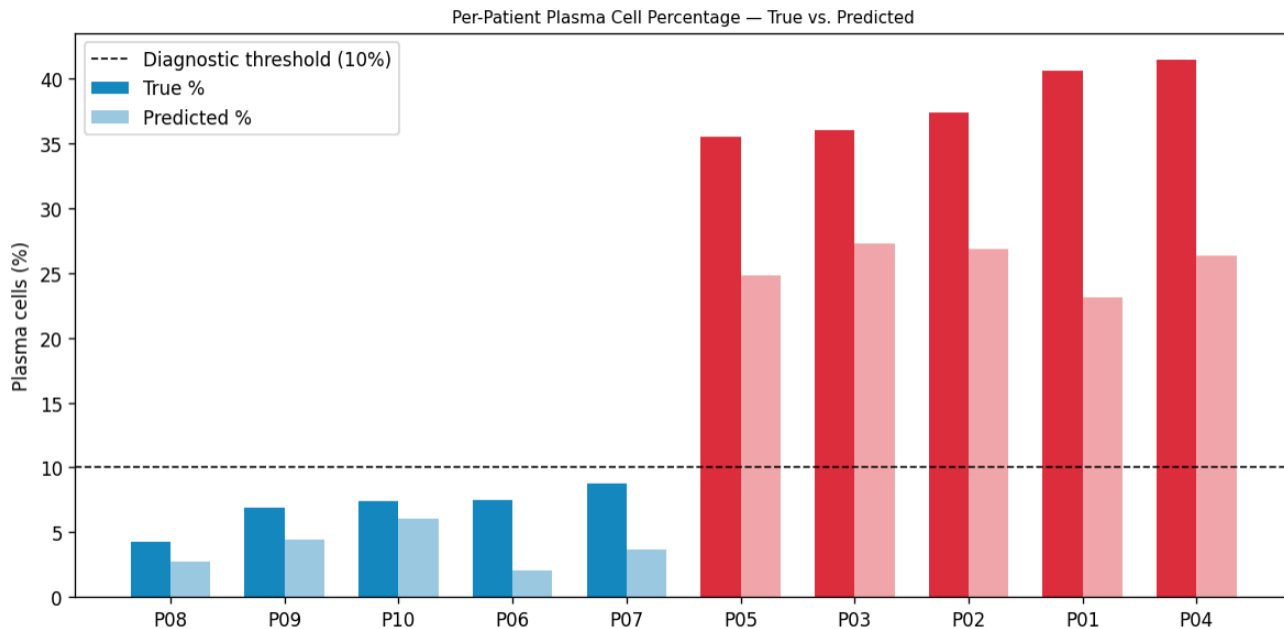
Patient 06 — **normal** true=8% · pred=2%



Both land on the correct side of the 10% threshold — patient-level diagnosis stays robust even when raw counts are under-predicted.

Diagnosis: true vs. predicted ratio

Every patient lands on the correct side of the 10% threshold — the diagnosis holds even when the predicted percentage is lower than the true one.



Let's code!

From cell counts to a patient diagnosis.

What the notebook does

-
- 01 Run the trained detector over **every slide** of each test patient
 - 02 Count plasma vs. non-plasma cells and **aggregate per patient**
 - 03 Compute the plasma-cell ratio and apply the 10% threshold
 - 04 Compare predicted diagnosis vs. expert ground truth per patient
-



Break

05

Synthetic data augmentation

Generating training cells that are invariant to how the slide was lit and captured. *Accepted, BRACIS 2026.*

What you will learn

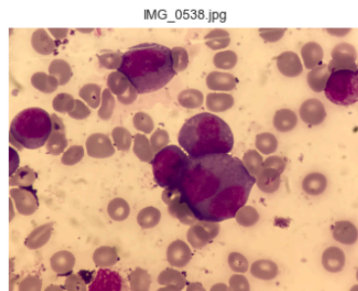
- 01 Estimate a slide's **background light** and other per-slide statistics
 - 02 Model those statistics with **KDE** to generate synthetic background templates
 - 03 Crop plasma cells from smears and isolate each with its **segmentation mask**
 - 04 Transform crops into a new color space (**color deconvolution**), using each slide's background light
 - 05 Train a generative model on those maps to synthesize new cells
-

TOPIC 01 • PER-SLIDE STATISTICS

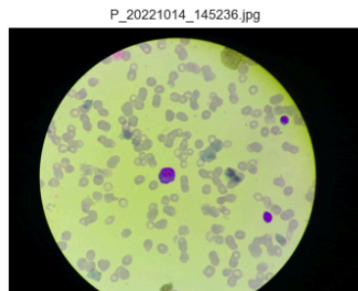
Background light estimation

Each slide has its own **background light** (the color of its unstained regions), which varies with staining intensity and illumination at capture.

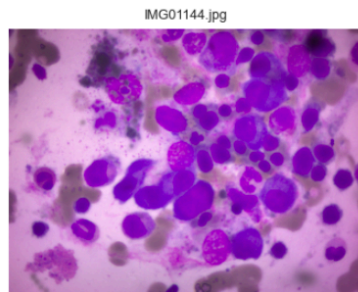
- 1 Downsample the slide to a **thumbnail**
- 2 Cluster pixel colors with **K-Means (k=5)**
- 3 Pick the **brightest centroid** — background is brighter than cells & debris



RGB: (243, 205, 157)



RGB: (201, 207, 136)



RGB: (217, 187, 230)



Learn distributions with KDE

Load the collected statistics, then fit **Kernel Density**

Estimation to learn approximate distributions of:

Cells per image

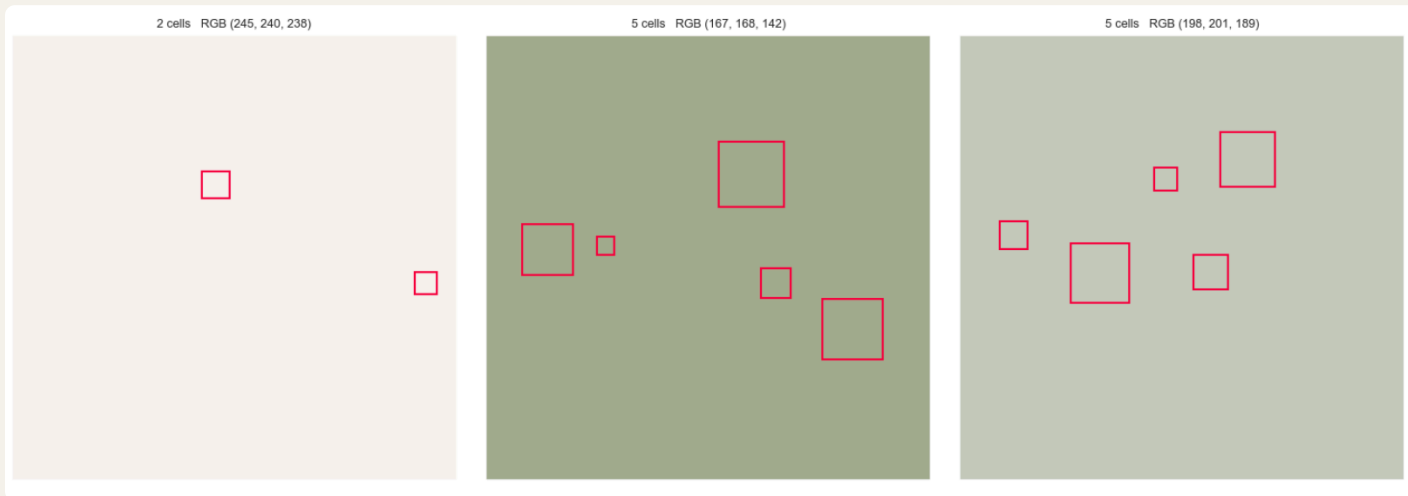
Background color

Cell positions

Cell sizes

Sampling them yields a synthetic **template** — a colored background plus positions where cells will be inserted.

SAMPLED TEMPLATES · background color + cell positions (red)



THE SKELETON

The template works as the **skeleton** of the synthetic image.

Background color + empty cell slots → generated cells get pasted in next.

TOPIC 03 • CROP & MASK THE CELL

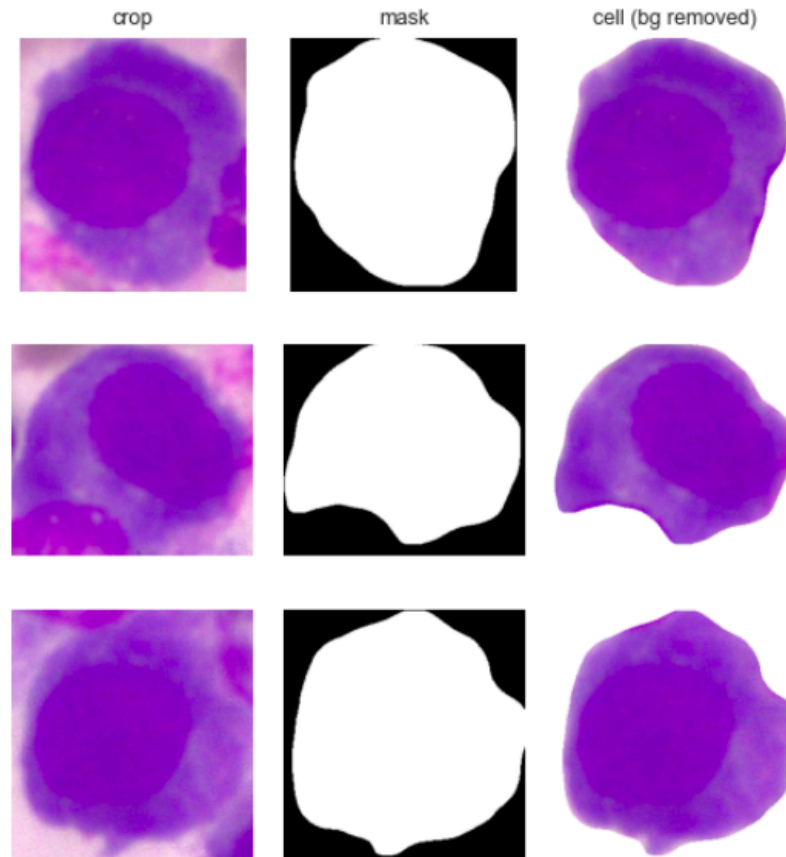
Isolate the real cell

Each real cell is **cropped** from the original image and isolated with a **binary mask**.

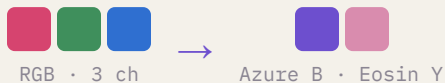
255 cell region

0 background

The mask **removes the surrounding background** — we work only with the cell itself.



From RGB to a stain space



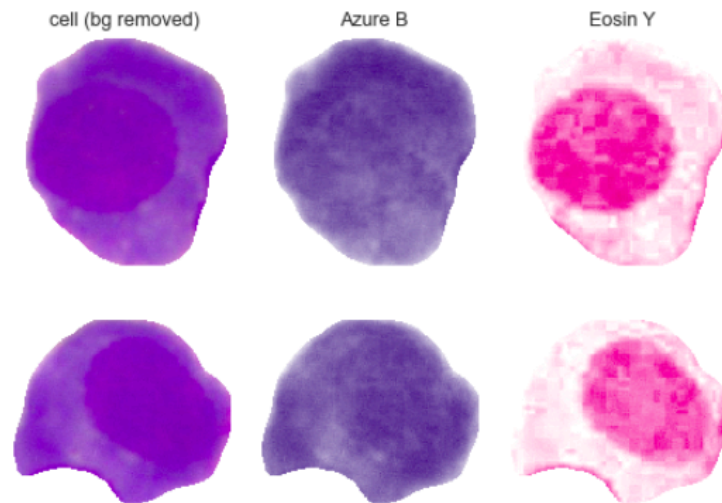
RGB mixes everything

"This cell has purple, pink and bluish tones." Biology and color are tangled together.

Dye space is structured

"More Azure B here, more Eosin Y there, almost no dye here" — controlled & interpretable.

Color deconvolution — IMG01117 IO = (239, 210, 236)



TOPIC 04 • BEER-LAMBERT LAW

Light absorbed $\approx$ dye present



Dark pixel $\rightarrow$ absorbed **much** light $\rightarrow$ likely **much** dye



Light pixel $\rightarrow$ absorbed **little** light $\rightarrow$ **little/no** dye

DECONVOLUTION

RGB $\rightarrow$ dye concentration

$\hookrightarrow$ **diffusion learns here**

INVERSE — RECONSTRUCT RGB

$$I = I_0 \cdot 10^{-(C \cdot S)}$$

Generate cells with diffusion

FORWARD real map → noise



Add Gaussian noise step by step until the map is pure noise.

REVERSE noise → new cell



A **UNet** predicts & removes the noise step by step until a clean map remains.

Training

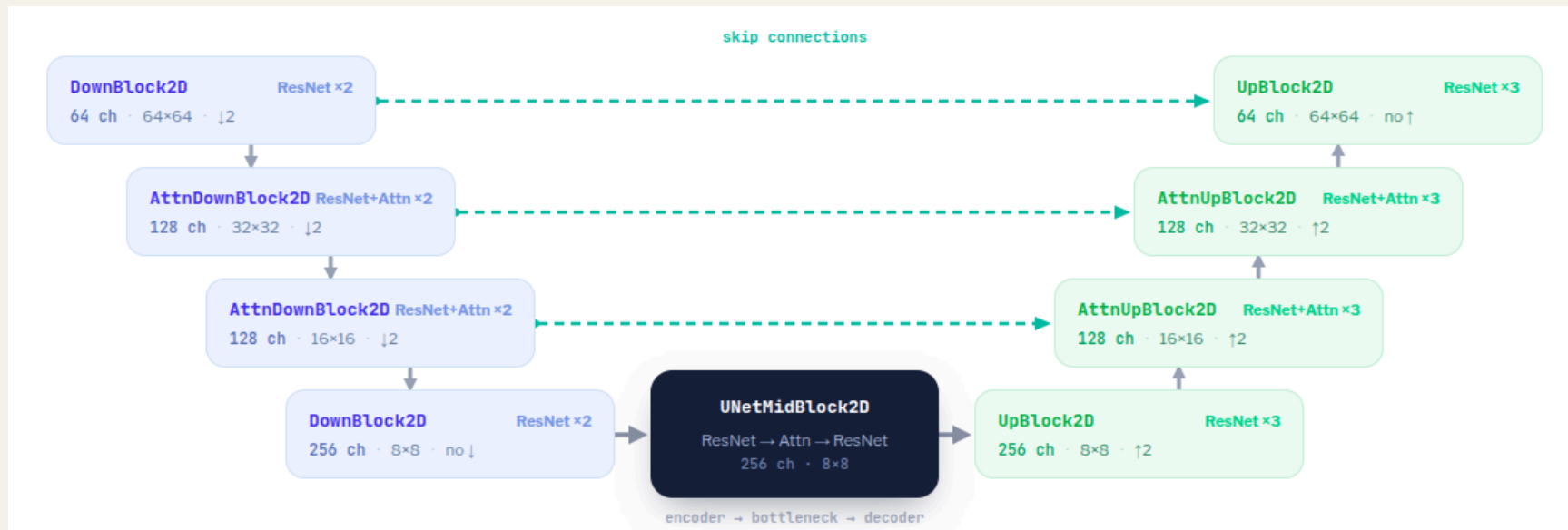
Pick a random step, add that much noise, train the UNet to predict it. **Loss = MSE** (predicted vs. true noise).

Data

Concentration maps from Part 6 — resized to 64×64, normalized with per-channel mean/std.

Encoder → bottleneck → decoder

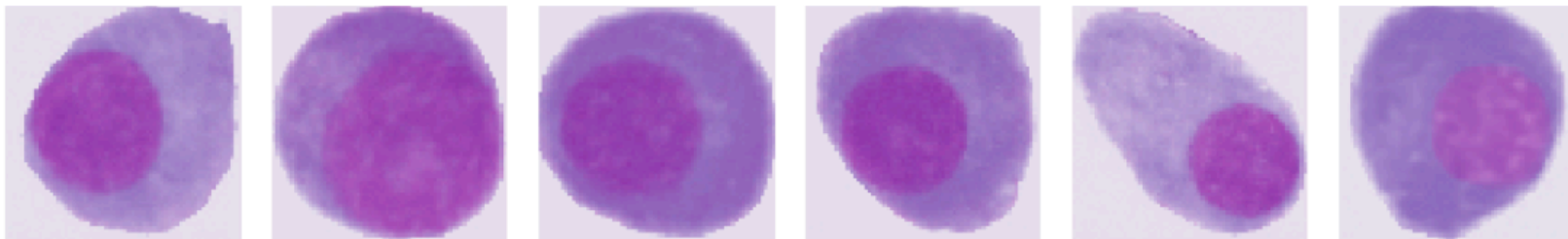
Downsampling encoder (64→128→256 ch) and a symmetric decoder, with **attention** at the middle resolutions and **skip connections** carrying fine detail across.



Cells generated by the model

Sampled from pure noise, then inverse-deconvolved to **RGB** under a near-white light. Each is newly generated — not a copy of any real cell.

Generated synthetic cells (RGB, near-white light)



RESULTS • PCMMD, PATIENT-WISE

+5.9

points mAP50-95

over the real-data baseline — **0.774 vs 0.715**, with DDPM at +40%.

15/15

augmented configs beat baseline

0.901

best mAP50 (DiT +30%)

TAKEAWAYS

What to remember

-
- 01 Deconvolving to **stain-concentration space** decouples morphology from illumination.
 - 02 Generating there + KDE templates **reliably lifts detection** in low-data regimes.
 - 03 **Fidelity $\neq$ utility**, and augmentation trades precision for recall — right for screening.
-

USED OUR DATASET AND CODE? DON'T FORGET CITING US!

BRACIS 2026	Batista Filho, J.L.S., Nogueira Rios, T., Araújo Rios, R.: Synthetic Data Augmentation for Plasmocyte Detection in Bone Marrow Images. In: Brazilian Conference on Intelligent Systems (BRACIS) (2026).
Scientific Data 2025	Andrade, C.L.B., Ferreira, M.V., Alencar, B.M., et al.: PCMMMD: A novel dataset of plasma cells to support the diagnosis of multiple myeloma. Scientific Data 12 (1), 161 (2025). doi.org/10.1038/s41597-025-04459-1
Scientific Reports 2024	Andrade, C.L.B., Ferreira, M.V., Alencar, B.M., et al.: Enhancing diagnostic accuracy of multiple myeloma through ML-driven analysis of hematological slides. Scientific Reports 14 (1), 11176 (2024). doi.org/10.1038/s41598-024-61420-9

Thank you!



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